

Computer science & IT

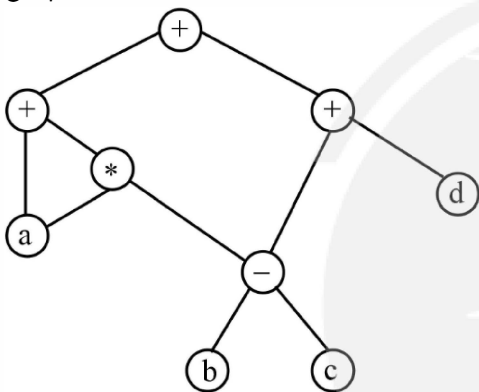
Compiler Design

DPP: 1

Intermediate code and optimization

Q1 The three addresses code involves_____.

- (A) At most 3 addresses.
- (B) Exactly 3 addresses.
- (C) At least 3 addresses.
- (D) No unary operator.

Q2 Consider the following DAG (Directed Acyclic graph):

Which of the following is the correct expression for given DAG?

- (A) $[a * a + (b - c) * (b - c) + d]$
- (B) $[a + a * (b - c) + (b - c) + d]$
- (C) $[a + a * (c - b) + (c - b) + d]$
- (D) $[a * a + (c - b) * (c - b) + d]$

Q3 Type checking is performed by:

- (A) Lexical analyses
- (B) Syntax analyses
- (C) Semantic analyses
- (D) Intermediate code generator

Q4 Which of the following are valid three addresses code (TAC)?

- (A) if $a > b$ goto C
- (B) return 0

(C) $x[i] = y;$ (D) $a = b * c$ **Q5** Which of the following is/are correct about abstract syntax tree (AST)?

- (A) AST is also used in program analysis and program transformation systems.
- (B) It is a tree representation of the abstract syntactic structure of source code written in a programming language.
- (C) It has no impact on the final output of the compiler.
- (D) It is the result of syntax analysis phase of a compiler.

Q6 Consider the following expression:

$$p + q * r + s - t - p + q * r$$

How many minimum number of temporary variable created in three addresses code?

Assume precedence order from lowest to highest is $-$, $+$, and $*$; and consider associativity of $+$ & $*$ is not important but $-$ is left associative.

Q7 Consider the following intermediate code:

1. $loc = -1$
2. $i = 0$
3. if $(i < 100)$ goto 5
4. goto 13
5. $t_1 = 4 * i$
6. goto 3
7. $t_2 = A[t_1]$
8. if $t_2 = x$ goto
9. goto 11
10. $loc = i$



11. $t_3 = i + 1$

12. $i = t_3$

13. goto 3

How many number of basic block from the given code?

Q8 Consider the following expression.

$((x * x) * (x * x) * ((x * x) * (x * x)))$

The total number of internal nodes in DAG representation are _____.

(A) 3 nodes and 5 edges

(B) 4 nodes and 7 edges

(C) 3 nodes and 6 edges

(D) 4 nodes and 6 edges



CUSTOMER
SERVICE

[Android App](#) | [iOS App](#) | [PW Website](#)

Answer Key

Q1 (A)**Q2 (B)****Q3 (C)****Q4 (A, B, C, D)****Q5 (A, B, D)****Q6 2****Q7 9****Q8 (D)**

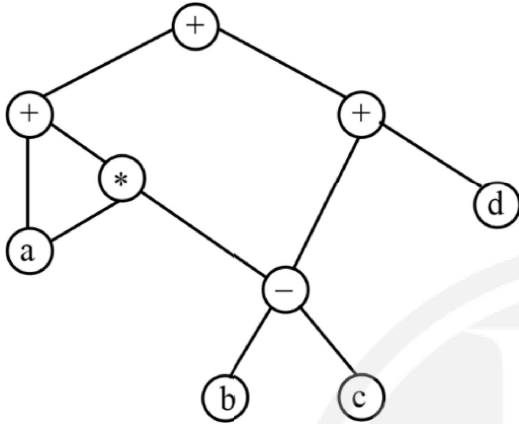
[Android App](#) | [iOS App](#) | [PW Website](#)

Hints & Solutions

Q1 Text Solution:

A three address code has at most three address location to calculate the expression.
Hence, option A is the correct answer.

Q2 Text Solution:



The correct expression is
 $a + a * (b - c) + (b - c) + d$
 So, option (b) is the correct answer.

Q3 Text Solution:

Semantic analysis mainly deals with the meaning of program and its execution. Type checking is an important aspect of semantic analysis where each operator should be compatible with its operands.

Q4 Text Solution:

A three addresses code instruction can have at most three operand. So, all of the options are correct.

Q5 Text Solution:

Abstract syntax tree has impact on the final output of the compiler. So, option (c) is incorrect else all options are correct.

Q6 Text Solution:

Given,
 $((p + (q * r) + s) - t) - (p + (q * r))$

So, three-address code for the expression is as follows:

$$t_1 = q * r;$$

$$t_2 = p * t_1;$$

$$t_1 = t_2 * t_1;$$

$$t_1 = t_1 * t_1;$$

$$t_1 = t_1 * t_2;$$

So, minimum 2 temporary variable are required.

Q7 Text Solution:

Basic block: It is the collection of three address code statements from leader to next leader without including the next leader is known as the basic block.

In the given three-addresses code, there are 9 leaders:

1, 3, 4, 5, 7, 9, 10, 11, 13

There are 9 basic blocks

Block B₁ : statements 1 – 2

Block B₂ : Statements 3

Block B₃ : Statements 4

Block B₄ : Statements 5 – 6

Block B₅ : Statements 7 – 8

Block B₆ : Statements 9

Block B₇ : Statements 10

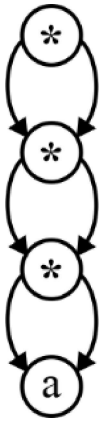
Block B₈ : Statements 11 – 12

Block B₉ : Statements 13

Q8 Text Solution:

DAG representation of given expression





4 nodes and 6 edges. So, option (d) is correct answer.

[Android App](#)

|

[iOS App](#)

|

[PW Website](#)